$$f(x_1, x_2 | w_1, w_2, w_0) = \underbrace{w_0 + w_1 x_1 + w_2 x_2}_{w^T \cdot x + w_0}$$

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = w_1$$

$$\frac{\partial f(x_1, x_2)}{\partial x_2} = w_2$$

where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \mathbf{w}$$

$$\frac{\partial w^T \cdot x}{\partial x} = w$$

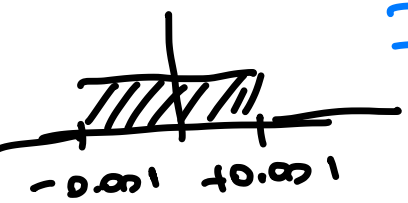
$$\frac{\partial w^T \cdot x}{\partial w} = x$$

$$\frac{\partial (w^T x + w_0)}{\partial w_0} = 1$$

Linear Discrimination for Binary Classification

STEP#1: Initialize $\{w, w_0\}$ randomly and decide η.

Initializing them to very small values; for example



Uniform$(-0.001, +0.001)$

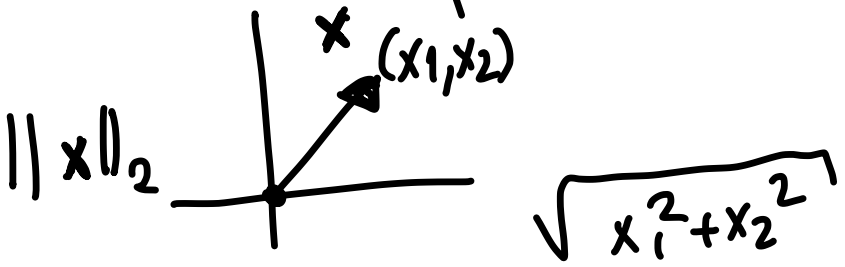
STEP# 2: Calculate Δw and Δw_0.

STEP#3: Update w and w_0.

$$w^{(t+1)} = w^{(t)} + \Delta w \longrightarrow -\eta \cdot \frac{\partial Error}{\partial w}$$

$$w_0^{(t+1)} = w_0^{(t)} + \Delta w_0 \longrightarrow -\eta \frac{\partial Error}{\partial w_0}$$

STEP#4: Go to STEP#2 if there is a change in the parameters. [i.e., $\|\Delta w\|_2 \neq 0$, $|\Delta w_0| \neq 0$]



Linear Discrimination (Multiple Classes) ($K > 2$)

$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$ $\quad x_i \in \mathbb{R}^D$ $\quad y_i \in \{1, 2, \dots, K\}$ → reference class

$$\sum_{c=1}^{K} Pr(y=c|x) = 1$$

$$Pr(y=c|x) \geq 0 \quad \forall c$$

$$\exp\left[\log\left[\frac{Pr(y=c|x)}{Pr(y=K|x)}\right]\right] = \exp\left[w_c^T x + w_{c0}\right]$$

$$\log\left[\frac{p(x|y=c)}{p(x|y=K)}\right] + \log\left[\frac{Pr(y=c)}{Pr(y=K)}\right]$$

$$Pr(y=1|x) + Pr(y=2|x) + \dots + Pr(y=K-1|x) + Pr(y=K|x) = 1$$

$$\frac{Pr(y=1|x) + Pr(y=2|x) + \dots + Pr(y=K-1|x)}{Pr(y=K|x)} = \frac{1 - Pr(y=K|x)}{Pr(y=K|x)}$$

$$\sum_{c=1}^{K-1} \frac{Pr(y=c|x)}{Pr(y=K|x)} = \frac{1}{Pr(y=K|x)} - 1$$

$$\frac{Pr(y=c|x)}{Pr(y=K|x)} = \exp\left[w_c^T x + w_{c0}\right]$$

$$Pr(y=K|x) = \frac{1}{1+\sum_{c=1}^{K-1} \exp[w_c^T x + w_{c0}]}$$

$$Pr(y=c|x) = \frac{\exp[w_c^T . x + w_{c0}]}{1+\sum_{d=1}^{K-1} \exp[w_d^T . x + w_{d0}]} \quad c=1,2,..,K-1$$

$\theta = \{ \underbrace{w_1}_{D\times 1}, \underbrace{w_{10}}_{1\times 1}, \underbrace{w_2}_{D\times 1}, \underbrace{w_{20}}_{1\times 1}, \dots, \underbrace{w_{(K-1)}}_{D\times 1}, \underbrace{w_{(K-1)0}}_{1\times 1} \}$ → total # of parameters $= (K-1)(D+1)$

Assume $w_K^T . x + w_{K0} = 0$

Softmax $\left[Pr(y=c|x) = \frac{\exp(w_c^T . x + w_{c0})}{\sum_{d=1}^{K} \exp(w_d^T x + w_{d0})} \right] \forall c$

$$g_1(x_{N+1}) = w_1^T x_{N+1} + w_{10}$$
$$\vdots$$
$$g_K(x_{N+1}) = w_K^T x_{N+1} + w_{K0}$$

} pick the one with maximum score.

IDEAL SCORES

1 +2 $\hat{Pr}(y=1|x) = \frac{\exp(g_1(x))}{\exp(g_1(x))+\exp(g_2(x))+\exp(g_3(x))}$ 0.7214

0 -2 $\hat{Pr}(y=2|x) = \frac{\exp(g_2(x))}{\exp(g_1(x))+\exp(g_2(x))+\exp(g_3(x))}$ 0.0132

0 +1 $\hat{Pr}(y=3|x) = \frac{\exp(g_3(x))}{\exp(g_1(x))+\exp(g_2(x))+\exp(g_3(x))}$ 0.2654

+20

-20

+10

0.9999

0.0007

0.0003

+2000

-2000

+1000

$$\frac{\exp(2000)}{\exp(2000)+\exp(-2000)+\exp(1000)} = \frac{\infty}{\infty}$$

$\exp(2000) \to \infty$, $\exp(-2000) \to 0$, $\exp(1000) \to \infty$

$$\frac{1}{1+\exp[-score]}$$

it is always SAFE.

$$\frac{\exp(2000)\ \exp(-2000)}{[\exp(2000)+\exp(-2000)+\exp(1000)]\ \exp(-2000)}$$

$$\frac{\exp(0)}{\exp(0)+\exp(-4000)+\exp(-1000)} = \frac{1}{1} = 1$$

$\exp(0) \to 1$, $\exp(-4000) \to 0$, $\exp(-1000) \to 0$

$$y_i|x_i \sim \text{Multinomial}\left(y_i;\ 1,\ \{\Pr(y_i=c|x)\}_{c=1}^{K}\right)$$



$$\left(\frac{1}{6}\right)^{1}\left(\frac{1}{6}\right)^{0}\left(\frac{1}{6}\right)^{0}\left(\frac{1}{6}\right)^{0}\left(\frac{1}{6}\right)^{0}$$

$$\text{Multinomial}\left(y_i;\ 1,\ \{1/6, 1/6, 1/6, \ldots, 1/6\}\right)$$

$$\text{likelihood}\left(\{w_c, w_{c0}\}_{c=1}^{K} \mid \mathcal{X}\right) = \prod_{i=1}^{N}\prod_{c=1}^{K} \Pr(y_i=c|x)^{\mathbb{1}(y_i=c)}$$

$$\mathbb{1}(y_i=c) \equiv y_{ic}$$

$$= \prod_{i=1}^{N}\prod_{c=1}^{K} \Pr(y_i=c|x)^{y_{ic}}$$

$$y = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 2 \end{bmatrix} \Rightarrow Y = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

one-hot encoding

$$\Downarrow$$

$$\hat{y}_{ic}$$

$$\log\left[\prod_{i=1}^{N}\prod_{c=1}^{K} \hat{y}_{ic}^{\,y_{ic}}\right]$$

$$\downarrow$$

$$\sum_{i=1}^{N}\sum_{c=1}^{K} y_{ic}\log(\hat{y}_{ic})$$

$$\text{loglikelihood} = \sum_{i=1}^{N}\sum_{c=1}^{K} y_{ic}\log\underbrace{\left[Pr(y_i=c|x_i)\right]}_{\hat{y}_{ic}}$$

$0.\log(0) \equiv 0$

$$\text{Error}(\{w_c, w_{c0}\}_{c=1}^{K} | \mathcal{X}) = -\sum_{i=1}^{N}\sum_{c=1}^{K} y_{ic}.\log(\hat{y}_{ic})$$

$$= -\sum_{i=1}^{N}\sum_{c=1}^{K} y_{ic}.\log\left[\frac{\exp\left[w_c^T.x_i + w_{c0}\right]}{\sum_{d=1}^{K}\exp\left[w_d^T.x_i + w_{d0}\right]}\right]$$

Exercise:

$$\frac{\partial \text{Error}}{\partial w_c} = ? \qquad \frac{\partial \text{Error}}{\partial w_{c0}} = ?$$

$$w_c^{(t+1)} = w_c^{(t)} + \Delta w_c$$

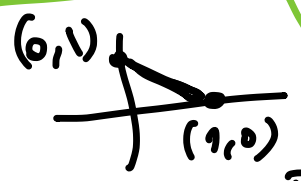
$$w_{c0}^{(t+1)} = w_{c0}^{(t)} + \Delta w_{c0}$$

$$\Delta w_d = \eta \sum_{i=1}^{N} \sum_{c=1}^{K} \frac{y_{ic}}{\sum_c y_{ic}} \cdot \hat{y}_{ic} \left[1(c=d) - \hat{y}_{id} \right] \cdot x_i$$

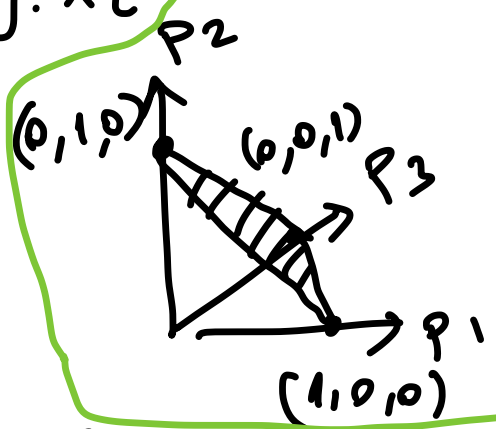
$$= \eta \sum_{i=1}^{N} \left[y_{id} - \hat{y}_{id} \right] \cdot x_i$$

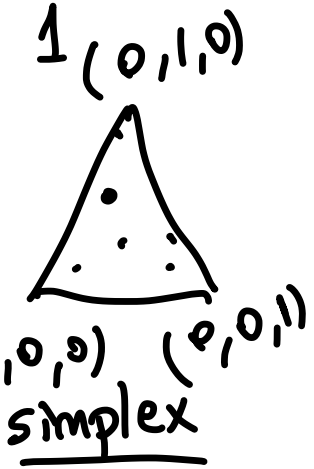
$$\Delta w_{d0} = \eta \sum_{i=1}^{N} \left[y_{id} - \hat{y}_{id} \right]$$

$p_1 + p_2 = 1$
$p_1 \geq 0$
$p_2 \geq 0$



$p_1 + p_2 + p_3 = 1$
$p_1 \geq 0$
$p_2 \geq 0$
$p_3 \geq 0$





simplex

ALGORITHM:

STEP #1: Initialize $\{w_1, w_{10}, w_2, w_{20}, \ldots, w_K, w_{K0}\}$ randomly to small values.

STEP #2: Calculate gradients

STEP #3: Update parameters using gradients

STEP #4: Go to STEP #2 if there is enough change in the parameters.